

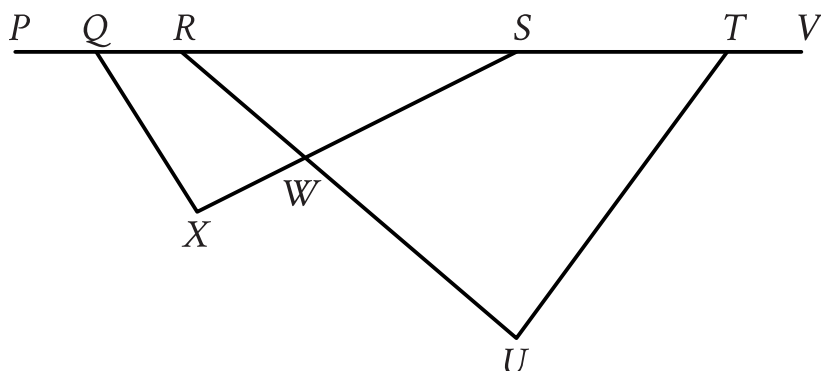
● HARD

● GEOMETRY AND TRIGONOMETRY

Lines, angles, and triangles

7 questions · ~11 min

03



Note: Figure not drawn to scale.

- From left to right, horizontal line segment upper P upper V includes the following points:
 - Upper P
 - Upper Q
 - Upper R
 - Upper S
 - Upper T
 - Upper V
- Points upper X and upper U are each below line segment upper P upper V.
- From left to right, the following 2 overlapping triangles are formed between points on line segment upper P upper V and points upper X and upper U:
 - Triangle upper Q upper X upper S
 - Triangle upper R upper U upper T
- Line segments upper R upper U and upper S upper X intersect at point upper W.
- A note indicates the figure is not drawn to scale.

In the figure shown, points Q , R , S , and T lie on line segment PV , and line segment RU intersects line segment SX at point W . The measure of $\angle SQX$ is 48° , the measure of

$\angle SXQ$ is 86° , the measure of $\angle SWU$ is 85° , and the measure of $\angle VTU$ is 162° . What is the measure, in degrees, of $\angle TUR$?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

GRID-IN

The area of triangle ABC above is at least 48 but no more than 60. If y is an integer, what is one possible value of x ?

●	/ ●	/ ●	●
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

LINES, ANGLES, AND TRIANGLES — SET 3 · GEOMETRY AND TRIGONOMETRY

Q3

GRID-IN GEOMETRY AND TRIGONOMETRY

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is $9x - 560^\circ$. The sum of the measures of one of the acute angles and three of the obtuse angles is $-18x + w^\circ$. What is the value of w ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN

GEOMETRY AND TRIGONOMETRY

In convex pentagon $ABCDE$, segment AB is parallel to segment DE . The measure of angle B is 139 degrees, and the measure of angle D is 174 degrees. What is the measure, in degrees, of angle C ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

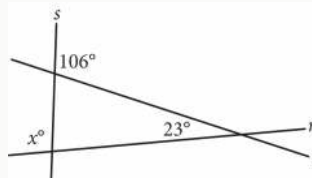
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

In triangles LMN and RST , angles L and R each have measure 60° , $LN = 10$, and $RT = 30$. Which additional piece of information is sufficient to prove that triangle LMN is similar to triangle RST ?

- A. $MN = 7$ and $ST = 7$
- B. $MN = 7$ and $ST = 21$
- C. The measures of angles M and S are 70° and 60° , respectively.
- D. The measures of angles M and T are 70° and 50° , respectively.

GRID-IN

Intersecting lines r, s, and t are shown below.



STUDENT-PRODUCED RESPONSE

○	/○	/○	○
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

LINES, ANGLES, AND TRIANGLES — SET 3 · GEOMETRY AND TRIGONOMETRY

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $7x - 250^\circ$. The sum of the measures of four of the eight angles is k° . Which of the following could NOT be equivalent to k , for all values of x ?

- A. $-14x + 1,540$
- B. $14x - 320$
- C. $-28x + 1,720$
- D. 360